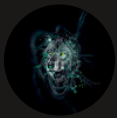




VPC Traffic Flow and Security



Kiprono Yegon

sg-02e7bcf5e0e4a261d - danSG

Security group (sg-02e7bcf5e0e4a261d | danSG) was created successfully

Details

sg-02e7bcf5e0e4a261d - danSG

Details

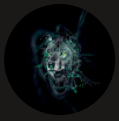
Security group name danSG	Security group ID sg-02e7bcf5e0e4a261d	Description A Security Group for the NextWork VPC.	VPC ID vpc-0500a63648f8de800
Owner 783259868137	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (1)

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-08c842bb2c5498924	IPv4	HTTP	TCP	80	0.0.0.0/0



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is an isolated cloud network and it is useful because it helps isolate resources and restrict the access

How I used Amazon VPC in this project

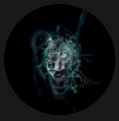
In today's project, I used Amazon VPC to create security groups and network Access Control Lists to control inbound and outbound traffic.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how using the EC2 global View simplifies the monitoring of resources that have been deployed in multiple regions

This project took me...

This project took me about an hour to complete



Route tables

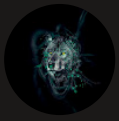
Route tables acts as a GPS for the resources in your subnet. They determine where data packets should go based on their destination IP address

Routes tables are needed to make a subnet public because they direct the internet-bound traffic to the internet gateway

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	local	Active	No	CreateRoute
	Internet Gateway	Active	No	CreateRoute
	igw-00334e756bcaecd1d	Active	No	CreateRoute

Add routeCancel



Route destination and target

Routes are defined by their destination and target, which means the destination is the IP address the data is to reach and the target is the path that the traffic will have to take to get to its destination.

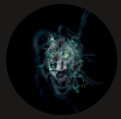
The route in my route table that directed internet-bound traffic to my internet gateway had a destination of 0.0.0.0/0 and a target of the internet gateway

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	local	Active	No	CreateRoute
	Internet Gateway			
	lgw-00334e756bcaecd1d			

Add route

Cancel



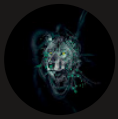
Security groups

Security groups are responsible for checking who comes in and out. They have strict rules about what kind of traffic can enter or leave the resource based on its IP address, protocols and port numbers. Every resource must be associated with a security group. This means security groups don't attach to a VPC or a subnet, they attach to a specific resource within that VPC/subnet. If you don't specify a security group when you launch a resource, it will use the default security group that AWS creates whenever you set up a VPC.

Inbound vs Outbound rules

Inbound rules control the data that can enter the resources in your security group, I configured an inbound rule that allows any IP address to access the public resources

Outbound rules are control that data that your resources can send out. By default, my security group's outbound rule allows all traffic to any destination



sg-02e7bcf5e0e4a261d - danSG

✔ Security group (sg-02e7bcf5e0e4a261d | danSG) was created successfully

▶ Details

sg-02e7bcf5e0e4a261d - danSG

Details

Security group name

danSG

Security group ID

sg-02e7bcf5e0e4a261d

Description

A Security Group for the NextWork VPC.

VPC ID

[vpc-0500a63648f8de800](#)

Owner

783259868137

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

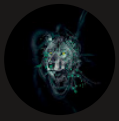
Tags

Inbound rules (1)

🕒 Manage tags

🔍 Search

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sgr-08c842bb2c5498924	IPv4	HTTP	TCP	80	0.0.0.0/0

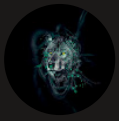


Network ACLs

Network ACLs are the entry point of the subnet, checking each data packet against a table of ACL rules before allowing them through.

Security groups vs. network ACLs

The difference between a security group and a network ACL is that network ACL's are used to set broad traffic rules that apply to an entire subnet while Security groups allow for more granular control, managing access to individual resource.



Default vs Custom Network ACLs

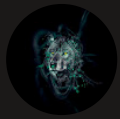
Similar to security groups, network ACLs use inbound and outbound rules

By default, a network ACL's inbound and outbound rules will allow all inbound and outbound traffic

In contrast, a custom ACL's inbound and outbound rules are automatically set to "Deny" - all inbound and outbound traffic are denied until you add rules about the kind of traffic you'll allow.

The screenshot shows the AWS Network ACL console for a custom ACL named 'acl-0e4d4c8df2f7aa072 / danNACL'. The 'Inbound rules' tab is selected, showing a table with 2 rules. The first rule (number 100) allows all traffic from 0.0.0.0/0. The second rule (number *) denies all traffic from 0.0.0.0/0.

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny



Tracking VPC Resources

I created additional VPC resources : VPC, Internet Gateway and security group. Instead of my usual region, I used a different region. Teams would use multiple regions to deploy resources to ensure high availability.

The EC2 Global View is a feature that lets you see and manage your EC2 and VPC resources across all AWS Regions from a single dashboard. Instead of switching between Regions to track your resources, this view gives you a single view at everything you have running, which is super useful for multi-region deployments. I could even narrow down my search by filtering using the region option. Without EC2 Global View, you'd have to manually switch and search in different regions.

Now that I've learnt about EC2 Global View, I'd use it again to view resources that I have created over multiple regions

The screenshot displays the AWS Global View interface. On the left, a sidebar contains links for 'Region Explorer', 'Global search', 'Regions and Zones', and 'Settings'. The main content area is titled 'AWS Global View' and includes a 'Summary' section with resource counts: Compute (0), Storage (2), Networking (368), and Security (52). Below this is the 'Region explorer (34)' section, which shows a table of resources across various AWS regions. The table has columns for Region, Instances, VPCs, Subnets, Security Groups, Volumes, Auto Scaling Groups, Route Tables, VPC Endpoints, and NAT Gateways. The regions listed include Asia Pacific (Mumbai), Asia Pacific (Osaka), Asia Pacific (Seoul), Asia Pacific (Singapore), Asia Pacific (Sydney), Asia Pacific (Tokyo), Canada (Central), Europe (Frankfurt), Europe (Ireland), Europe (London), and Europe (Paris).

Region	Instances	VPCs	Subnets	Security Groups	Volumes	Auto Scaling Gr...	Route Tables	VPC Endpoints	NAT G
Asia Pacific (Mumbai) ap-south-1	-	1	3	1	-	-	1	-	-
Asia Pacific (Osaka) ap-northeast-3	-	1	3	1	-	-	1	-	-
Asia Pacific (Seoul) ap-northeast-2	-	1	4	1	-	-	1	-	-
Asia Pacific (Singapore) ap-southeast-1	-	1	3	1	-	-	1	-	-
Asia Pacific (Sydney) ap-southeast-2	-	1	3	1	-	-	1	-	-
Asia Pacific (Tokyo) ap-northeast-1	-	1	3	1	-	-	-	-	-
Canada (Central) ca-central-1	-	1	3	1	-	-	1	-	-
Europe (Frankfurt) eu-central-1	-	1	3	1	-	-	1	-	-
Europe (Ireland) eu-west-1	-	1	3	1	-	-	1	-	-
Europe (London) eu-west-2	-	1	3	1	-	-	1	-	-
Europe (Paris) eu-west-3	-	1	3	1	-	-	1	-	-



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